**Cambridge International Examinations**
Cambridge International General Certificate of Secondary Education

CANDIDATE NAME

CENTRE NUMBER | CANDIDATE NUMBER

*5527988714*

**ENVIRONMENTAL MANAGEMENT** **0680/21**

Paper 2 **May/June 2018**

**1 hour 45 minutes**

Candidates answer on the Question Paper.

No Additional Materials are required.

**READ THESE INSTRUCTIONS FIRST**

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **both** questions.

Electronic calculators may be used.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of **15** printed pages and **1** blank page.

DC (NH/FC) 144329/4


CAMBRIDGE International Examinations

[Turn over

1 **(a)** The table shows climate data for the city of Ndola in Zambia.

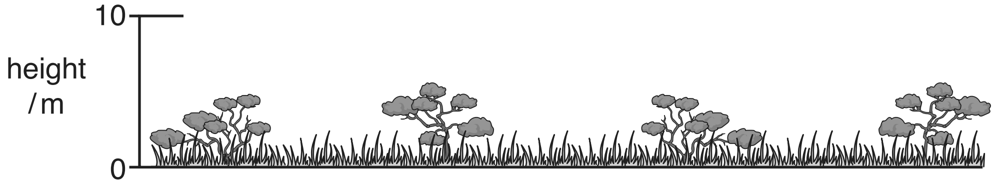
| | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **mean monthly temp /°C** | 20.8 | 20.8 | 21.0 | 20.5 | 18.6 | 16.5 | 16.7 | 19.2 | 22.5 | 23.7 | 22.5 | 21.0 |
| **mean monthly rainfall /mm** | 293 | 249 | 170 | 46 | 4 | 1 | 0 | 0 | 3 | 32 | 130 | 306 |

Use the table to complete the following paragraph.

The wettest month is ............................., which has a mean monthly rainfall of ............................. mm. Temperatures are highest in ............................. and the annual range of temperature is ............... °C. Ndola has a savanna climate and is located in the ............................. hemisphere. [5]

**(b)** The diagram shows two types of savanna vegetation.


tree savanna




shrub savanna

**(i)** Describe **two** differences between the two types of savanna vegetation.

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[2]

(ii) State the type of vegetation between the trees in the tree savanna.

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(iii) Explain how desertification can occur in savanna regions.

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(c) The sentences give five definitions.

**A** A diagram of energy flows among species in an ecosystem.

**B** The area or type of environment in which a particular kind of animal or plant usually lives.

**C** The process by which green plants and some other organisms use sunlight to synthesise glucose from carbon dioxide and water.

**D** When individuals or seeds move from one site to a breeding or growing site.

**E** An interaction between organisms that require the same limited resource.

Match the terms to the definitions, **A**, **B**, **C**, **D** and **E**.

| **term** | **letter** |
|---|---|
| dispersal | .................... |
| competition | .................... |
| food web | .................... |
| habitat | .................... |
| photosynthesis | .................... |

[4]

**(d)** Some environments have been affected by tourism. The photograph shows part of a tourist resort.

**(i)** Suggest why this was a good location to build a tourist resort.

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**(ii)** Suggest the environmental impacts of building a resort such as this one.

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**(iii)** The resort faces environmental challenges:

- a shortage of water – the average daily use is 880 litres per tourist and there are 30 000 swimming pools in the area
- a massive amount of waste and sewage to process.

Describe how these environmental challenges can be managed.

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**(iv)** The resort receives 1.75 million tourists per year, at least half from other countries.

Explain how this tourism will cause air pollution.

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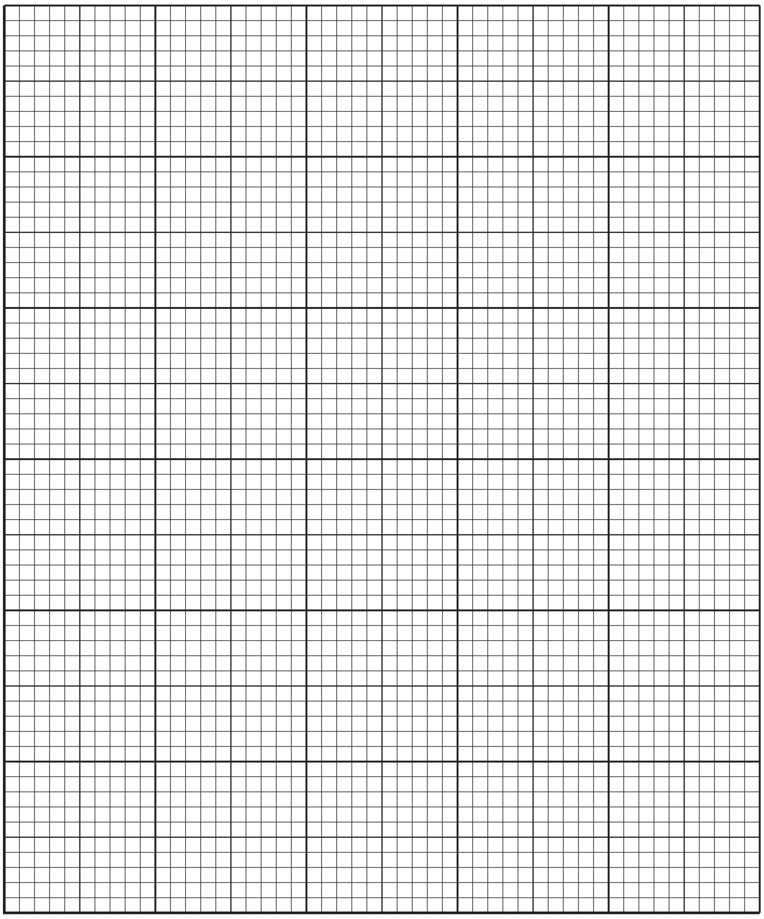
**(e)** The table shows the number of tourists travelling abroad from 2000 to 2014.

| year | number of tourists /million |
|---|---|
| 2000 | 680 |
| 2004 | 730 |
| 2008 | 900 |
| 2012 | 1040 |
| 2014 | 1130 |

Draw a line graph on the grid to show this data.



[4]

**(f)** 'Tourism is good for the economy, but bad for the planet.'

How far do you agree with this statement? Give reasons for your answer.

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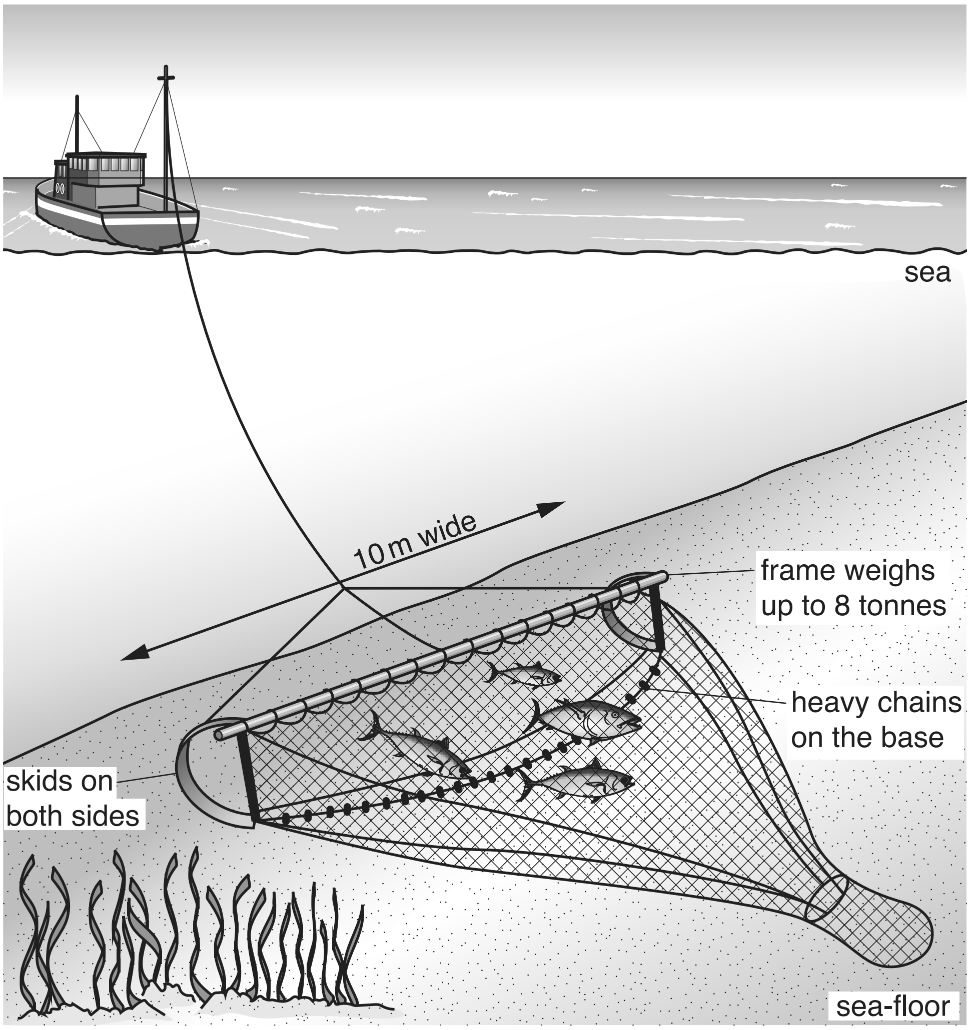
**2** **(a)** The diagram shows a trawler and its net (trawl).




not to scale

**(i)** Describe how fish are caught by a trawler.

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**(ii)** Describe how trawling can damage the sea-floor.

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**(iii)** What is meant by the term *overfishing*?

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**(iv)** Explain why overfishing has occurred in many of the world's oceans.

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**(v)** Suggest how overfishing can impact a marine food web.

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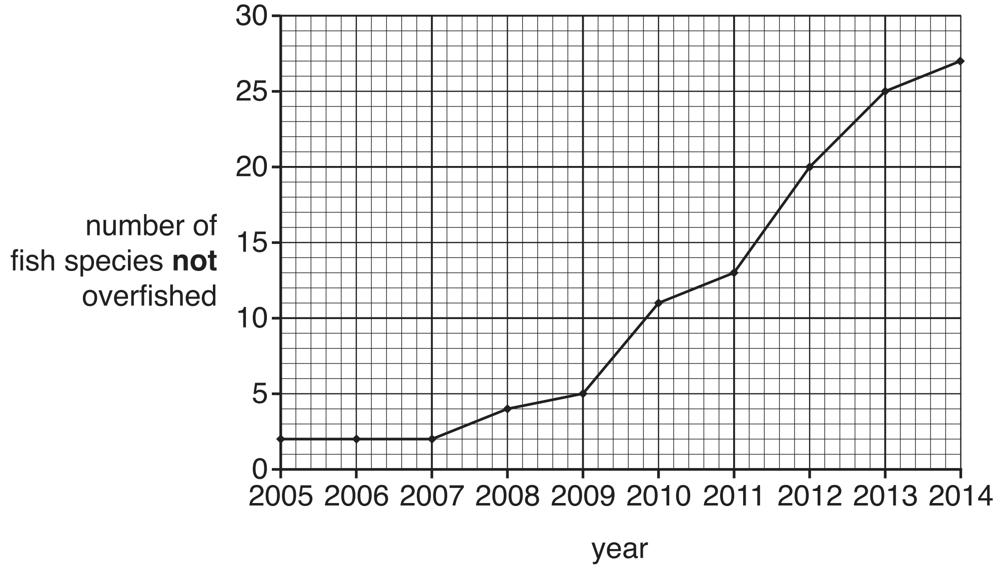
**(b)** The graph shows the number of fish species that are **not** overfished in the north east Atlantic Ocean and North Sea from 2005 to 2014.




**(i)** State the number of fish species that are **not** overfished in 2012.

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**(ii)** Describe what the graph shows about the changes in fish species that are **not** overfished from 2005 to 2014.

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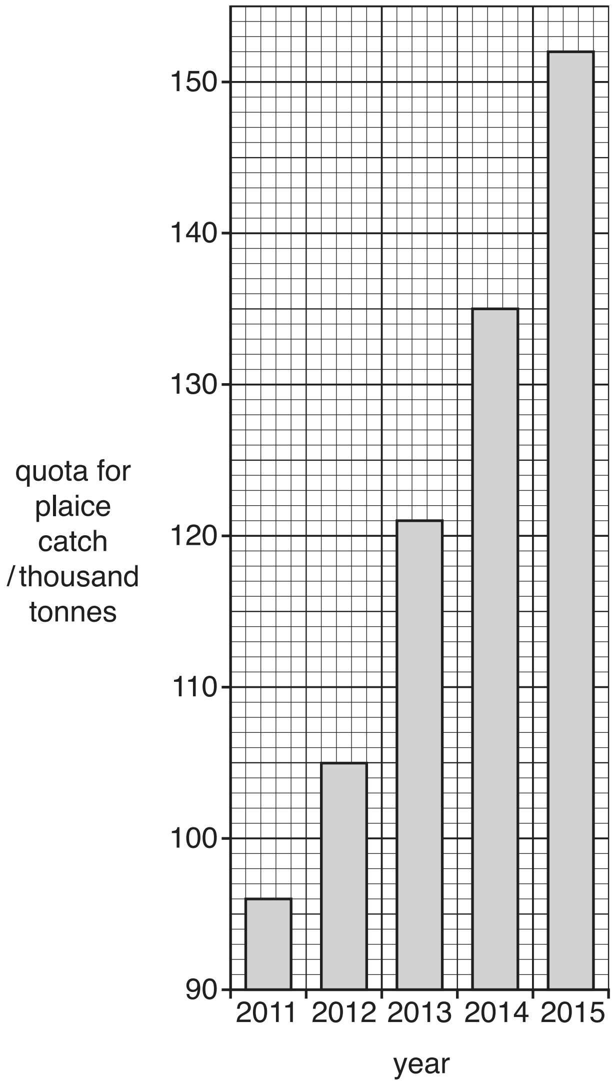
**(c)** The graph shows the quotas for the plaice catch in the north east Atlantic Ocean and North Sea from 2011 to 2015. Plaice are a species of fish.



**(i)** State the quota for the plaice catch in 2015.

................................ thousand tonnes [1]

**(ii)** Calculate the increase in the quota for the plaice catch from 2011 to 2015.

Show your working.

............................... thousand tonnes [2]

**(iii)** Suggest why governments have increased the quota for the plaice catch since 2011.

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**(iv)** Describe **three** ways in which fish stocks can be managed to reduce overfishing, other than by using quotas.

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**(d)** The table shows information about tropical cyclones (hurricanes) in the Atlantic Ocean and Caribbean Sea from 2000 to 2009.

| **year** | **number of cyclones** | **approximate number of deaths** | **cost of damage /billion USD** |
|---|---|---|---|
| 2000 | 8 | 79 | 1.2 |
| 2001 | 9 | 105 | 7.1 |
| 2002 | 4 | 23 | 2.6 |
| 2003 | 7 | 92 | 4.4 |
| 2004 | 9 | 3100 | 50.0 |
| 2005 | 15 | 2280 | 159.0 |
| 2006 | 5 | 14 | 0.5 |
| 2007 | 6 | 423 | 3.0 |
| 2008 | 8 | 1047 | 42.0 |
| 2009 | 3 | 6 | 77.0 |

**(i)** State the year with the lowest cost of damage.

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**(ii)** State the three-year period with the most cyclones.

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**(iii)** Calculate the average number of cyclones per year for this ten-year period.

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**(iv)** Suggest why some cyclones caused more damage than other cyclones.

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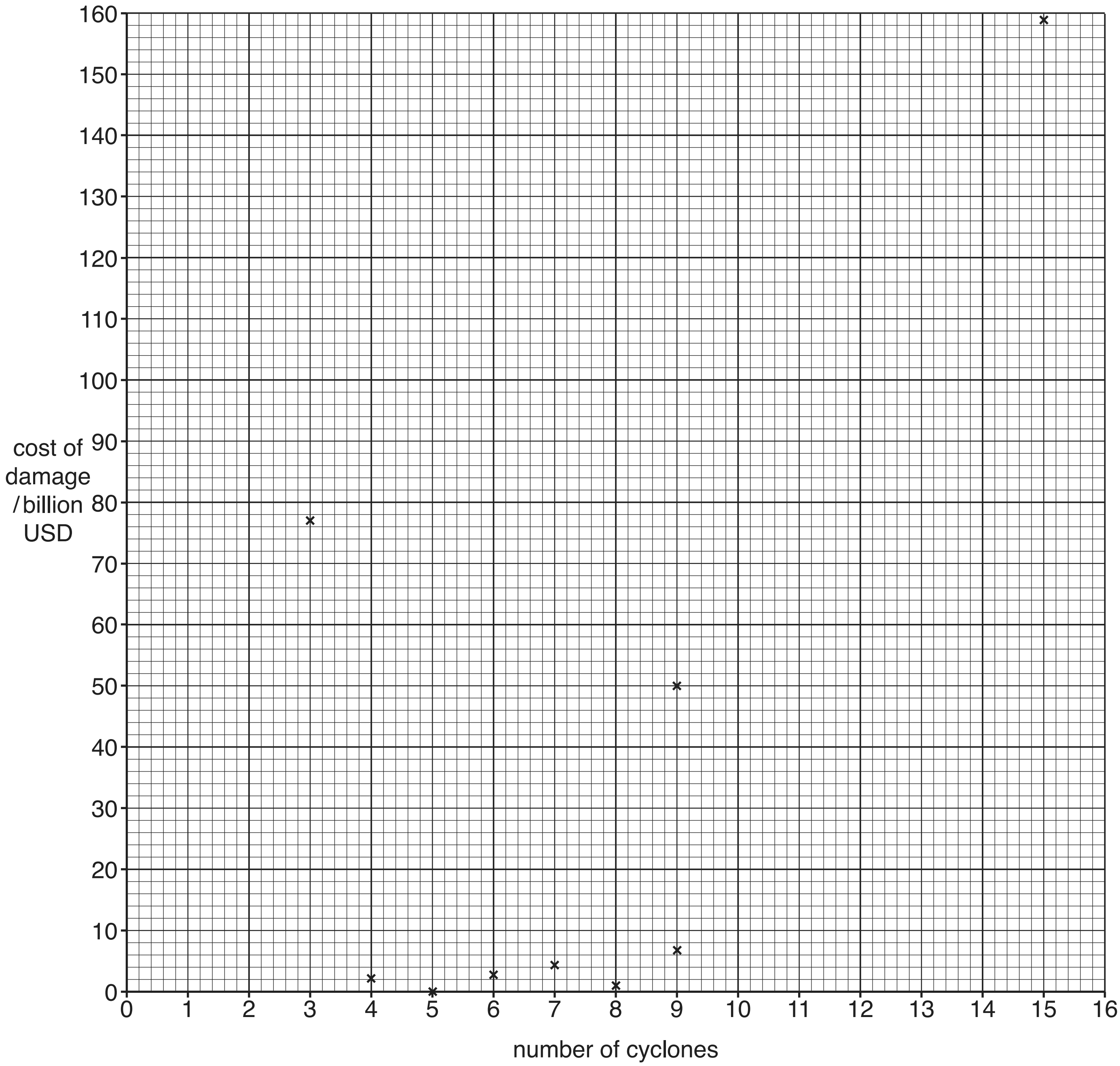
**(v)** The graph shows the number of cyclones and the cost of damage in billion USD.

Complete the graph, by adding the data for 2008 from the table in **(d)**.




[1]

**(vi)** Is there a relationship between the number of cyclones and the cost of damage? Justify your answer.

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**(vii)** Describe the causes of cyclones.

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**(e)** Is it possible to reduce the pollution in the oceans? Explain your answer.

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